

Interwoven Cultures of Nablus

Input file: standard input
Output file: standard output
Time limit: 2 seconds
Memory limit: 1024 megabytes



Nablus's winding streets and lively markets reflect a city that has flourished as a commercial and cultural center for over 2,000 years. Offering Aamer a glimpse into its history, he discovered how Nablus has been considered the main commercial and industrial center in Palestine, and even well-known for its sweets like Nabulsi Kunafa.

You are given a string S of n Latin characters, and an array A of n integers. Each element has a value of A_i . Find the length of the longest good substring.

A substring $S[l, r]$ ($1 \leq l \leq r \leq n$) is considered good if both of the conditions below are true:

1. $S[l, r]$ is a palindrome ($S_{l+i} = S_{r-i}$ for all $0 \leq i \leq r-l$)
2. $A_l | A_{l+1} | \dots | A_{r-1} | A_r = \sum_{i=l}^r A_i$, where $|$ is the bitwise OR operator.

Noot: $x|y$ means x bitwise or y .

Input

The first line contains the number of test cases ($1 \leq t \leq 10^3$).

The first line of each test case contains n ($1 \leq n \leq 10^5$).

The second line of each test case contains a string S of length n .

The last line of each test case contains n space-separated integers $A_1, A_2, \dots, A_n$ ($1 \leq A_i \leq 10^9$) — the array A .

It is guaranteed that the sum of n over all test cases does not exceed $3 \cdot 10^5$.

Output

For each test case, print one integer:

The maximum length of a substring that fulfills both of the conditions above.

Example

standard input	standard output
3	2
4	1
aaaa	3
1 2 3 4	
1	
a	
10	
6	
abbabb	
1 2 3 4 8 7	